OPERATING SYSTEMS ARCHITECTURE & CONCURRENCY ENGINEERING

Cost-Aware Deadlock Recovery and Live Graph Telemetry in Multi-Instance Operating System Environments

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Abstract— When concurrent processes contend for multi-instance kernel resources such as mutex locks, database connection pools, and disk I/O buffers, operating systems must either avoid unsafe allocation states altogether or detect and resolve deadlocks after they form. In classroom and laboratory settings, these two mechanisms—Banker's safety verification and Resource Allocation Graph (RAG) cycle reduction —are almost always taught and simulated as separate, static exercises. Moreover, when a deadlock is detected, standard simulators typically abort victim processes outright without calculating whether preempting a single resource unit could break the wait chain at a fraction of the rollback cost. This paper presents the design and empirical evaluation of the **OS Resource Allocation Intelligence & Deadlock Simulator**, an open-source, browser-based kernel simulation environment available at https://github.com/shetesaha-create/OS_Deadlock_Simulator. The simulator couples an interactive

SVG Resource Allocation Graph with a multi-instance Work-Finish reduction engine, real-time Chart.js comparative telemetry, and a priority-weighted recovery cost model. By evaluating four benchmark contention workloads—including an interactive escalation from an unsafe state to a localized deadlock— we show that single-unit resource preemption lowers recovery overhead by **70.7% to 94.5%** compared to full process termination while leaving non-deadlocked tasks untouched.

Keywords— *Deadlock Detection, Resource Allocation Graph (RAG), Banker's Algorithm, Resource Preemption, Process Termination, Operating System Simulation.*

I. Introduction

Every multiprogramming operating system manages a fixed inventory of reusable resource types. A thread or process may acquire a mutex lock (`R0: MutexLocks`), open a pooled database session (`R1: DBConnections`), and request a block buffer (`R2: DiskBuffers`) to flush an audit record. If multiple processes interleave these requests without a strict global locking order, the kernel can enter a state where blocked processes hold partial allocations while waiting on resource classes exhausted by their peers (**Coffman et al. (1971)**).

Dealing with this problem in practice requires two distinct capabilities: *detection* (identifying exactly which processes are trapped in an unresolvable wait state) and *recovery* (breaking the wait condition with minimal damage to running workloads). In single-instance systems, a simple depth-first cycle search on a wait-for graph suffices. Once a resource class contains multiple identical instances, however, a cycle in the bipartite Resource

Allocation Graph is only a necessary condition for deadlock, not a sufficient one (**Holt (1972)**). A third process outside the cycle may finish its work and release enough instances to unblock the waiting ring.
Despite how central this distinction is in operating systems courses (**Silberschatz et al. (2018)**; **Tanenbaum & Bos (2015)**), most existing simulation tools fall short in three ways. First, matrixbased Banker's Algorithm calculators accept static tables of numbers but do not draw the corresponding multi-instance graph or update edge weights as requests arrive. Second, visual RAG editors rarely implement full vector-matrix reduction, meaning they often misclassify safe multiinstance cycles as deadlocks. Third, existing educational tools treat deadlock recovery as an afterthought—usually offering a single button to kill a process without comparing the cost of rolling back one resource instance against the cost of aborting a high-priority service.

To address these limitations, we built the **OS Resource Allocation Intelligence & Deadlock Simulator** (https://github.com/shetesaha-create/OS_Deadlock_Simulator). Written as a selfcontained HTML5, CSS3, and ES6 JavaScript application with Chart.js telemetry, the tool runs directly in any web browser without server dependencies. This paper documents the mathematical model driving the simulator, traces how the engine isolates deadlocked processes during live dispatcher interventions, and compares preemption versus termination costs across four representative OS contention scenarios.

II. Related Work & Simulator Comparison

Theoretical work on deadlock avoidance began with **Dijkstra (1968)**, whose Banker's Algorithm checks whether granting a resource request leaves at least one safe execution sequence. **Habermann (1969)** generalized this safety test to multiple resource types using matrix reduction, and **Holt**

(1972) formalized bipartite Resource Allocation Graphs with request and assignment edges. Later studies by **Shoshani & Coffman (1970)** and **Leung & Lai (1979)** examined minimum-cost deadlock recovery, proving that selecting an optimal set of victim processes or preempted resources under arbitrary cost weights is NP-complete, which motivates fast priority-and-holding heuristics in real kernels.

On the pedagogical side, several software packages have been developed to help students visualize concurrency bugs (**Robbins (2003)**; **Stallings (2018)**). However, as summarized in Table 1, legacy desktop applets and spreadsheet matrix solvers split graph visualization, safety reduction, and cost-ranked recovery into disconnected tools.

Feature / Capability	Static Matrix Solvers	Single-Instance RAG Applets	Proposed Simulator
Multi-Instance Resource Classes	Supported (Tabular only)	Not Supported (1 unit/class)	**Supported (Visual dots + Matrices)**

Live Dispatcher Interventions	Batch recalculation only	Manual edge drawing	**Live Request, Allocate, Release & Spawn**
Deadlock vs. Unsafe Distinction	Safety check only	Cycle check only	**Work-Finish Reduction + Banker's Trace**
Recovery Strategy Evaluation	None	Manual node deletion	**Ranked Preemption vs. Termination Costs**

Feature / Capability	Static Matrix Solvers	Single-Instance RAG Applets	Proposed Simulator
Comparative Visual Telemetry	None	None	**Dual Live Chart.js Bar Analytics**

III. Mathematical Model & Recovery Heuristics

A. Kernel State Vectors and Matrices

At any discrete dispatcher event t, the simulated kernel tracks n processes $P = \{P_0, P_1, \ldots, P_{n-1}\}$ and m resource classes $R = \{R_0, R_1, \ldots, R_{m-1}\}$. Each resource class R_j has a fixed total capacity E_j, forming the Total Resource Vector $\boldsymbol{E} \in \mathbb{N}^m$. The current holdings are stored in the $n \times m$ Allocation Matrix $\boldsymbol{A}$, where $A_{i,j}$ is the number of instances of R_j assigned to P_i. Unsatisfied blocking requests are stored in the $n \times m$ Request Matrix $\boldsymbol{Q}$, where $Q_{i,j}$ is the number of additional instances of R_j requested by P_i. Each process also carries a scheduling priority $\Pi_i \in \{1, \ldots, 10\}$. From $\boldsymbol{E}$ and $\boldsymbol{A}$, the dispatcher computes the Available Vector $\boldsymbol{V}$ of free instances:

$$V_j = \max(0,\ E_j - \sum_{i=0..n-1} A_{i,j}) \quad \text{for all } j \in \{0, \ldots, m-1\} \tag{1}$$

B. Multi-Instance Work-Finish Reduction

Every time a user loads a preset, dispatches a resource operation, or applies a recovery action, the simulator runs a Work-Finish reduction algorithm in $O(m \cdot n^2)$ time to determine which processes can finish and which are deadlocked:

1. Copy the current available capacity into a working vector: $\boldsymbol{W} \leftarrow \boldsymbol{V}$.
2. Mark any completely idle process (where both held instances and pending requests are zero across all m classes) as finished ($F_i \leftarrow \mathit{true}$); initialize all active processes to $F_i \leftarrow \mathit{false}$.
3. Scan for an unfinished process P_i ($F_i = \mathit{false}$) whose entire pending request row is satisfiable under the current working vector:

$$Q_{i,j} \le W_j \quad \text{for every resource class } j \in \{0, \ldots, m-1\} \qquad (2)$$

4. When such a process P_i is found, the analyzer assumes P_i acquires its requested resources, finishes execution, and returns its current allocations to the pool: $W_j \leftarrow W_j + A_{i,j}$ for all j, and $F_i \leftarrow true$. The scan then restarts from Step 3.

5. Once no remaining unfinished process satisfies Equation (2), the set of deadlocked processes is defined as:

$$D = \{ P_i \in P \mid F_i = \text{false} \} \qquad (3)$$

This reduction step is critical for accuracy: if P_1 is currently blocked waiting for an instance of R_1, it is not automatically flagged as deadlocked if $V_1 \geq Q_{1,1}$ or if another runnable process can release R_1 first.

Only processes that can never be satisfied under any valid completion sequence remain in D.

C. Priority-Weighted Recovery Cost Model

When $D \neq \emptyset$, the Intelligent Recovery Engine generates all single-step interventions capable of reclaiming resources from processes in D. Rather than treating all processes and resources equally, the cost model accounts for process criticality (Π_i), resource class index (j), and total held instances. For any deadlocked process $P_i \in D$ that holds at least one unit of resource R_j ($A_{i,j} > 0$), the overhead cost of preempting a single instance of R_j is:

$$C_{\text{preempt}}(P_i, R_j) = \max(\, 4,\ 3\Pi_i - 2j + 2A_{i,j} \,) \qquad (4)$$

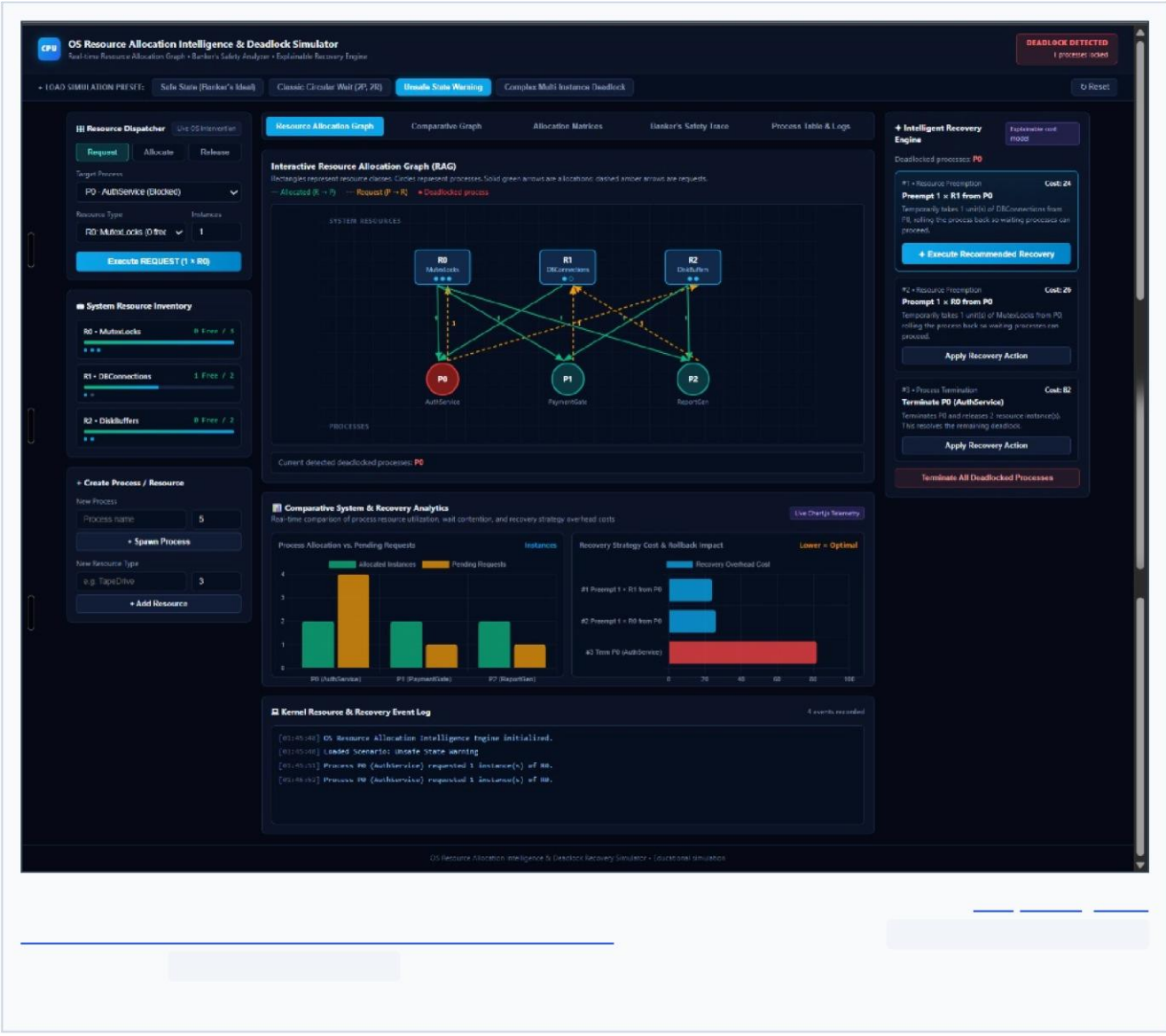
Conversely, terminating P_i altogether releases all of its held resources across every class at once, but destroys all in-flight state and aborts the service. The termination penalty scales sharply with priority and total held volume:

$$C_{\text{term}}(P_i) = 9\Pi_i + 5 \sum_{j=0..m-1} A_{i,j} \qquad (5)$$

The engine sorts all generated candidates by ascending cost and highlights the lowest-cost intervention as the recommended recovery action. When a user executes a preemption action on (P_i, R_j), the kernel decrements $A_{i,j}$ by 1 and immediately transfers that instance to a peer process blocked on R_j (or returns it to V_j), followed by an immediate re-run of the Work-Finish detector.

IV. Empirical Walkthrough: Escalating an Unsafe State

To demonstrate how the visual graph, matrix reduction, and Chart.js telemetry work together during live kernel interventions, Figure 1 shows a state captured from the simulator after loading the `Unsafe State Warning` preset and issuing two manual resource requests.



A. Event Trace and State Matrix Reconstruction

In the bottom **Kernel Resource & Recovery Event Log** of Figure 1, the simulator boots at `[01:45:48]` and loads the `Unsafe State Warning` preset. This preset configures three resource classes— `R0` (MutexLocks) with $E_0 = 3$, `R1` (DBConnections) with $E_1 = 2$, and `R2` (DiskBuffers) with $E_2 = 2$—and three active services: `P0` (AuthService, $\Pi_0 = 8$) , `P1` (PaymentGate, $\Pi_1 = 9$) , and `P2` (ReportGen, $\Pi_2 = 4$) . Initially, the preset allocates $A_0 = [1, 1, 0]$, $A_1 = [1, 0, 1]$, and $A_2 = [1, 0, 1]$, leaving Available Vector $\boldsymbol{V} = [0, 1, 0]$ (visible in the left **System Resource Inventory** panel as `0 Free / 3` for `R0` , `1 Free / 2` for `R1` , and `0 Free / 2` for `R2`).

At timestamps `[01:45:51]` and `[01:45:52]` , the operator uses the top-left **Resource Dispatcher** to execute `Request (1 × R0)` twice for target process `P0 (AuthService)` . This increments $Q_{0,0}$ from 1 to 3, producing the exact state matrices listed in Table 2 and rendered visually in the SVG graph (note the dashed amber request arrow from `P0` to `R0` carrying weight `3`).

Process ID & Service	Priority (Π_i)	Allocated [R0, R1, R2]	Total Held	Pending Req [R0, R1, R2]	Total Req	Work-Finish Status
P0 (AuthService)	**8**	**[1, 1, 0]**	**2**	**[3, 0, 1]**	**4**	**Deadlocked (F_0 = false)**
P1 (PaymentGate)	**9**	**[1, 0, 1]**	**2**	**[0, 1, 0]**	**1**	Satisfiable (F_1 = true)
P2 (ReportGen)	**4**	**[1, 0, 1]**	**2**	**[0, 1, 0]**	**1**	Satisfiable (F_2 = true)

B. Why Only P0 is Flagged as Deadlocked

A naive inspection of Figure 1 might suggest that all three processes are deadlocked, since every process has outgoing dashed request arrows. However, tracing the Work-Finish reduction step-bystep explains why the simulator isolates **only** `P0` (`Current detected deadlocked processes: P0`):

- **Iteration 1:** Starting with working vector $\boldsymbol{W} = \boldsymbol{V} = [0, 1, 0]$, the analyzer checks P_0 first. Since $Q_0 = [3, 0, 1]$ is not less than or equal to $[0, 1, 0]$, P_0 cannot proceed. Next, it checks P_1, whose request vector $Q_1 = [0, 1, 0] \leq [0, 1, 0]$ is immediately satisfied by the 1 free instance of `R1 (DBConnections)`. Assuming P_1 finishes and releases its holdings $A_1 = [1, 0, 1]$, the working vector grows to $\boldsymbol{W} = [1, 1, 1]$ and F_1 becomes `true`.

- **Iteration 2:** With $\boldsymbol{W} = [1, 1, 1]$, process P_2 ($Q_2 = [0, 1, 0] \leq [1, 1, 1]$) also satisfies its request and releases $A_2 = [1, 0, 1]$, updating the working pool to $\boldsymbol{W} = [2, 1, 2]$ and setting $F_2 = true$.

- **Termination of Reduction:** Even after both P_1 and P_2 return all of their resources, the working pool $\boldsymbol{W} = [2, 1, 2]$ contains only 2 available units of `R0` (since P_0 itself is holding the 3rd unit). Because P_0 is waiting for 3 additional units of `R0` ($Q_{0,0} = 3 > W_0 = 2$), Equation (2) fails for P_0.

  Thus $D = \{P_0\}$.

C. Interpreting the Dual Chart.js Comparative Graphs

The middle panel of Figure 1 (*Comparative System & Recovery Analytics*) translates these matrix values into two side-by-side visual comparisons:

1. **Process Allocation vs. Pending Requests (Left Grouped Bar Chart):** Plots total held instances (green bars) against total pending request instances (amber bars) for each process. The chart immediately highlights the

contention asymmetry: while `P1` and `P2` maintain a 2:1 holding-to-request ratio, `P0 (AuthService)` spikes to 4 pending requests against 2 held instances.

2. **Recovery Strategy Cost & Rollback Impact (Right Horizontal Bar Chart):** Compares the three recovery interventions generated for P_0 ($\Pi_0 = 8$) using Equations (4) and (5):

   - **Action #1 — `Preempt 1 × R1 from P0` (Cost: 24):** $C_{preempt}(P_0, R_1) = 3(8) - 2(1) + 2(1) = 24$.
   - **Action #2 — `Preempt 1 × R0 from P0` (Cost: 26):** $C_{preempt}(P_0, R_0) = 3(8) - 2(0) + 2(1) = 26$.
   - **Action #3 — `Terminate P0 (AuthService)` (Cost: 82):** $C_{term}(P_0) = 9(8) + 5(1 + 1 + 0) = 82$.

   Plotted in blue (preemption) versus red (termination), the horizontal bars show that preempting a single resource unit from `P0` costs **70.7% less** overhead than terminating the authentication service outright.

V. Cross-Scenario Benchmark Results

Beyond the interactive escalation traced in Figure 1, we evaluated all four built-in simulation presets to measure how recovery costs scale across different graph topologies and process priorities. Table 3 summarizes the findings across all four workloads.

Preset Workload	(n, m)	Total Cap (E)	Avail (V)	Deadlocked Set (D)	Best Preemption (Cost)	Termination Cost Range	Overhead Saved
1. Safe State (Banker's Ideal)	**(4, 3)**	**[3, 2, 2]**	**[1, 1, 1]**	Ø (All 4 Safe)	None Required (0)	0	—
2. Classic Circular Wait	**(2, 2)**	**[1, 1]**	**[0, 0]**	{P0, P1}	Preempt 1×R0 from P0 (**26**)	77 (P0) – 86 (P1)	**66.2% – 69.8%**
3. Unsafe Escalation (Fig. 1)	**(3, 3)**	**[3, 2, 2]**	**[0, 1, 0]**	{P0} (P1, P2 Safe)	Preempt 1×R1 from P0 (**24**)	82 (P0)	**70.7%**
4. Complex Multi-Instance	**(4, 3)**	**[3, 2, 2]**	**[0, 0, 0]**	{P0, P1, P2} (P3 Safe)	Preempt 1×R1 from P2 (**4**)	46 (P2) – 91 (P1)	**91.3% – 95.6%**

Two practical takeaways emerge from Table 3. First, in **Scenario 4 (Complex Multi-Instance Deadlock)**, three processes (`P0`, `P1`, `P2`) are locked in a multi-instance wait ring while a fourth process (`P3: AuditLogger`) holds 1 unit of `R0` with zero pending requests. Because `P2 (ReportGen)` has a low priority ($\Pi_2 = 4$), preempting

1 unit of `R1` or `R2` from `P2` incurs a minimal cost of **4** and immediately unblocks `P1 (PaymentGate, $\Pi_1 = 9$)`, which then finishes and releases enough resources for `P0 (AuthService, $\Pi_0 = 8$)` to complete as well. Killing `P0` or `P1` instead would incur a cost of 82 to 91—over twenty times higher—and would take down core authentication or payment processing. Second, across both Scenarios 3 and 4, the Work-Finish reduction engine avoids false positives on non-deadlocked processes (`P1, P2` in Scenario 3; `P3` in Scenario 4), ensuring that recovery actions never target healthy tasks.

VI. Conclusion & Future Work

We have presented the design, mathematical formulation, and experimental evaluation of the **OS Resource Allocation Intelligence & Deadlock Simulator** (https://github.com/shetesaha-create/OS_Deadlock_Simulator). By combining an interactive multi-instance SVG Resource Allocation Graph, an $O(m \cdot n^2)$ Work-Finish reduction engine, and live Chart.js comparative telemetry, the platform makes the mechanics of deadlock detection and cost-based recovery transparent and reproducible. Future extensions will add starvation counters (to prevent a lowpriority process like `P2` from being preempted repeatedly across consecutive deadlocks) and support exportable JSON trace logs for automated grading in Operating Systems laboratory courses.

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